

IPL DATA ANALYSIS

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Project Name : AI Agent Observability Dashboard using Power BI

Project Guide : Ms. Rajshri Thete

Objective

The objective of the AI Agent Observability Dashboard is to analyze AI agent execution data and transform raw operational records into meaningful business insights using Microsoft Excel. The project focuses on monitoring AI agent performance, execution quality, response latency, token consumption, memory utilization, planner efficiency, failure patterns, and user satisfaction through an interactive dashboard.

Exploratory Analysis

- Understanding AI agent execution performance across different AI models.
- Comparing successful and failed executions.
- Analyzing response latency and execution efficiency.
- Evaluating token consumption across AI models.

Data Visualization

- Representing AI agent statistics using KPI Cards, Bar Charts, Clustered Column Charts and Tables.
- Creating interactive dashboards for easy analysis.
- Comparing Successful Executions through visual reports.

Performance Analysis

- Total AI Agent Executions
- Memory-assisted Executions
- Average Response Latency
- User Satisfaction
- Token Consumption

Dashboard Development

- Excel Tables,Pivot Tables
- Pivot Charts,Calculated Columns
- Excel Formulas,Conditional Formatting
- Dynamic KPI Cards,Slicers,Dashboard Design
- nteractive Filtering,Professional Dashboard Formatting

Problem Statement

Modern AI systems generate a large volume of execution logs containing information about model performance, response quality, latency, token usage, retrieval effectiveness, planner strategies, memory utilization, and execution failures.

Analyzing this information manually is difficult because of the large number of execution records and multiple performance indicators.

This project aims to use Microsoft Excel to transform AI execution data into an interactive Business Intelligence dashboard by addressing the following challenges:

1. Importing and preprocessing AI execution datasets while ensuring the data is clean, accurate, and suitable for analysis.
2. Monitoring AI performance using execution quality, latency, memory utilization, token usage, and user satisfaction metrics.
3. Identifying execution failures and operational bottlenecks using interactive visualizations.

Data Description

Source

The dataset consists of AI Agent Observability execution records containing operational information generated during AI agent interactions.

Attribute Name	Description
Run_ID	Unique execution identifier
Timestamp	Date and time of execution
Model	AI model used
Planner	Planning strategy used
Domain	Application domain
Query Length	User query length
Memory Enabled	Indicates whether memory was enabled
Input Tokens	Tokens received from the user
Output Tokens	Tokens generated by the AI
Total Tokens	Total tokens consumed
Tool Count	Number of external tool calls
Reasoning Steps	Number of reasoning iterations
Estimated Cost (USD)	Estimated execution cost
Retrieval Hits	Number of retrieved documents
Retrieval Quality Score	Quality of retrieved information
Context Relevance Score	Relevance of retrieved context
Answer Quality Score	Overall answer quality
User Satisfaction Score	User rating after execution
Latency (Seconds)	Response time
Final Success	Execution status (Success/Failure)
Failure Type	Type of execution failure

Data Model

The dataset follows a tabular structure where each row represents one AI agent execution.

1. Data Model Overview

AI Model,Planner Type,Domain ,Memory Usage,Token Consumption, Execution Cost, Retrieval Quality ,Answer Quality ,Context Relevance ,User Satisfaction,Failure Information ,Response Latency .

Entities and Attributes

2.Data Relationships

- Every execution belongs to one AI Model.
- Every execution belongs to one Planner Type.
- Every execution belongs to one Domain.
- Every execution has one execution status.
- Every execution consumes input and output tokens.
- Every execution records latency and estimated cost.
- Every execution generates quality scores.
- Failed executions are linked with their respective failure types.

Approach

1. Data Import

Software Used

- Microsoft Excel
- Pivot Tables
- Pivot Charts

Process

- Imported AI execution dataset.
- Converted raw data into an Excel Table.

- Verified column names and data types.
- Prepared the dataset for analysis.

2. Data Cleaning

- Checked for missing values.
- Replaced blank values where required.
- Removed duplicate records. Standardized categorical values.
- Corrected inconsistent data, Verified Boolean fields.
- Created helper columns, Categorized execution cost.
- Categorized latency, Categorized user satisfaction.
- Prepared the dataset for Pivot Tables.

3. Data Processing

Additional calculated columns were created including:

- Success Status, Memory Status
- Cost Category, Latency Category
- Satisfaction Category, Quality Level
- Hour Extraction, Token Categories

These calculated fields improved reporting and dashboard analysis.

4. Dashboard Development

An interactive dashboard was created using:

- KPI Cards, Pivot Charts
- Slicers, Dynamic Filters
- Professional Formatting
- Color Themes, Icons
- Interactive Visualizations

All dashboard components update automatically based on slicer selections.

Project Result

1. Data Preparation

The dataset was successfully cleaned, validated, and transformed into an analysis-ready format.

2. Executive KPI Dashboard

Displays:

- Total Runs, Successful Runs
- Memory Success, Average User Satisfaction
- Average Response Latency

Result:

Provides an overall operational overview of AI system performance.

3. Domain Analysis

Displays:

- Average Retrieval Quality
- Average Estimated Cost
- Average Latency

Result:

Identifies which business domains achieve better AI performance.

4. Failure Analysis

Displays:

- Failure Type Distribution

Result:

Helps identify the most common execution failures and operational issues.

5. AI Model Performance

Displays:

- Average Answer Quality
- Average User Satisfaction
- Average Estimated Cost
- Average Retrieval Quality

Result:

Compares the performance of different AI models and identifies the most effective model.

6. Token Usage Analysis

Displays:

- Average Prompt Tokens
- Average Output Tokens
- Average Total Tokens

Result:

Shows token consumption across AI models and supports cost optimization.

7. Cost Performance

Displays:

- Low Cost Executions
- Medium Cost Executions
- High Cost Executions

Result:

Provides insights into AI execution cost distribution.

8. Retrieval Quality Trend

Displays:

- Retrieval Quality Score

Result:

Evaluates retrieval effectiveness and highlights opportunities for improving response quality.

9. Interactive Filtering

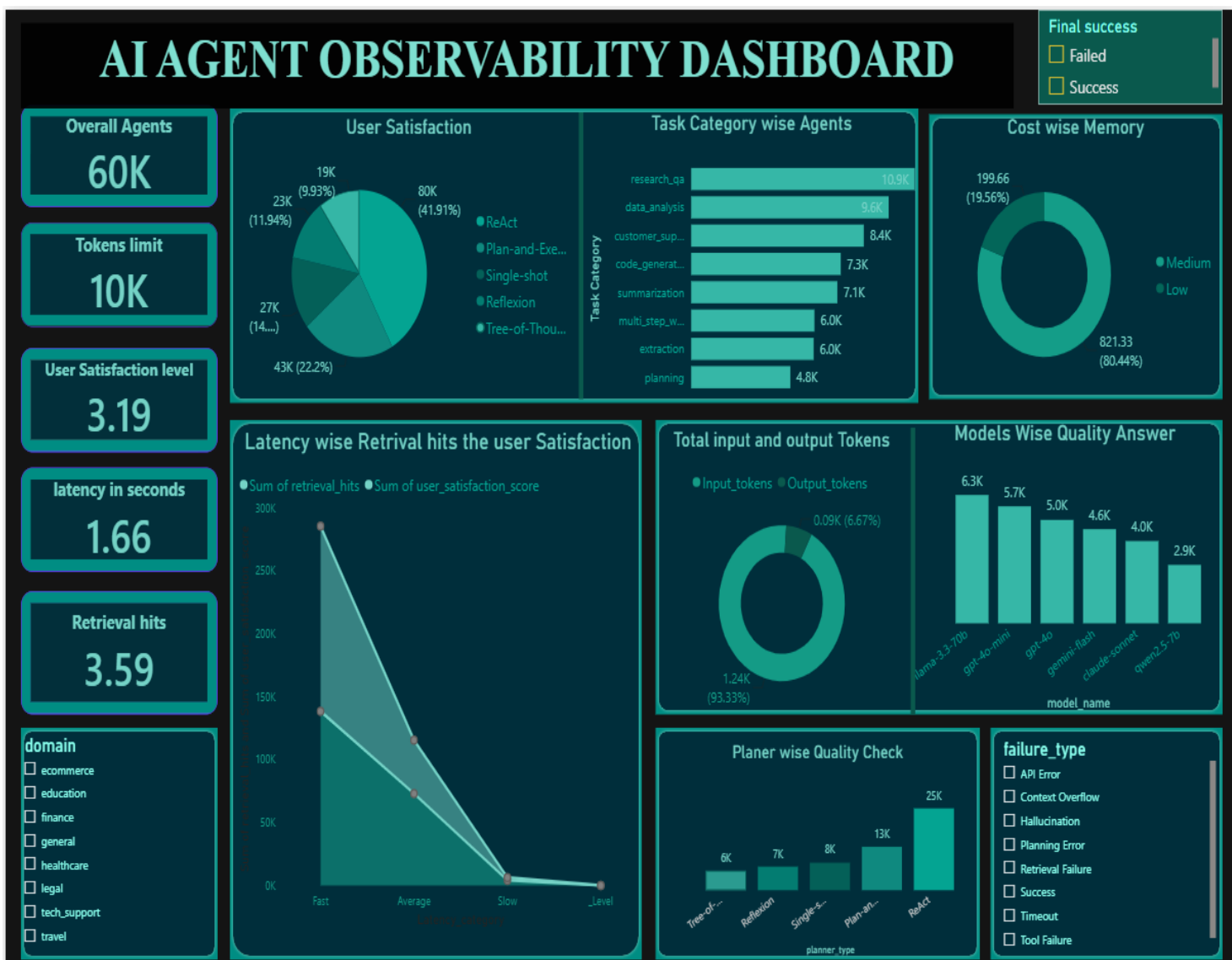
Dashboard includes slicers for:

- AI Model,Domain
- Planner Type
- Final Success
- Failure Type

Result:

Enables dynamic filtering and interactive business analysis.

Dashboard Screenshot :



Conclusion

The AI Agent Observability Dashboard successfully transforms raw AI execution data into a professional, interactive Business Intelligence solution using Microsoft Excel.

The dashboard provides comprehensive insights into AI model performance, execution quality, latency, token utilization, memory usage, retrieval effectiveness, operational costs, and user satisfaction. Through dynamic KPI cards, Pivot Charts, slicers, and interactive filtering, decision-makers can quickly evaluate system health, compare AI models, identify operational bottlenecks, and monitor key performance indicators.

The project demonstrates the practical application of Microsoft Excel for Business Intelligence by integrating data cleaning, calculated columns, Pivot Tables, Pivot Charts, dynamic dashboards, and advanced Excel features to support data-driven decision-making.

Future Enhancements

- Integration with live AI execution logs.
- Automated dashboard refresh using Power Query.
- Predictive analysis of execution failures using machine learning.
- Real-time monitoring with Power BI integration.
- AI Health Score based on multiple performance metrics.
- Trend analysis using historical execution data.
- Automated alerting for high latency, high cost, or increased failure rates.
- Executive reporting with printable and shareable dashboard summaries.